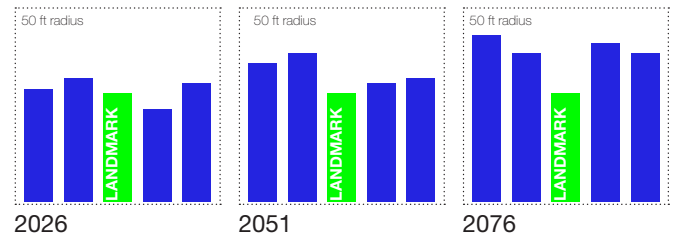


FIXED LANDMARKS, MOVING CITY FABRIC - MAHLON HOWARD

01_ATTEMPTING

How the height relationship between individual landmarks and their surrounding building fabric might change over time. Assignment 2 measured each landmark's height against the median height of buildings within a 500-foot radius. This forecast asks what that relationship could look like in the future if surrounding buildings change while landmark buildings remain fixed.



02_DATASET

- 1) NYC Building Footprints dataset
- 2) Designated and Calendared Buildings and Sites

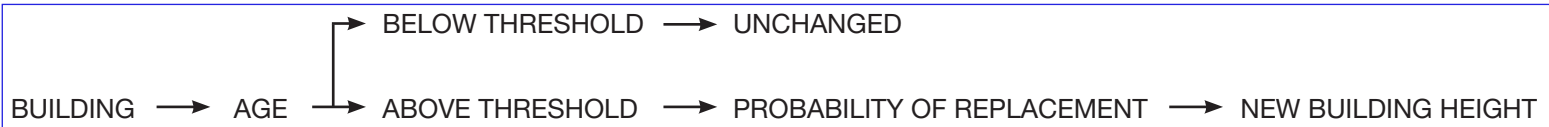
<u>COLUMN</u>	<u>MEANING</u>	<u>UNIT</u>
bin	Building Identification Number	ID
height_roof	Roof height	feet
construction_year	Construction year	year
geometry	Building footprint	geometry
base_bbl	Borough/Block/Lot identifier	ID

03_RESEARCH

- Bradley & Kohler (2007), "Methodology for the survival analysis of urban building stocks."
<https://www.tandfonline.com/doi/full/10.1080/09613210701266939>
At each time step, each building has a probability of surviving based on its age. Buildings that do not survive are removed from the stock.
- Hu et al. (2019), "Estimating Lifetimes and Stock Turnover Dynamics of Urban Residential Buildings in China."
<https://www.mdpi.com/2071-1050/11/13/3720>
It models new construction, aging, and demolition, with demolition treated as a stochastic process based on a Weibull hazard function calibrated from historical building-stock data.

05_RULE

At each time step, buildings older than a specified lifespan have a probability of being replaced. Replacement buildings receive a new height based on the height distribution of recently constructed buildings in their surrounding area. Individual landmark buildings remain unchanged.



06_RUN

2026 - 2031 - 2036 - 2041 - 2046 - 2051

Question: Does the landmark become increasingly exceptional as its surroundings change, or does its surrounding fabric eventually approach it in height?

07_DATA GENERATED

The simulation generates future building heights, building ages, demolition/replacement events, and the changing height difference between individual landmarks and their surrounding 500-foot urban fabric.

08_CAN'T SEE

The forecast describes what happens under the model's rules, not what will necessarily happen in Manhattan.

- Zoning changes
- Economic conditions
- Real estate development decisions
- Individual preservation decisions
- Changes in landmark designation